

Exp 26: Transaction Level Modeling (TLM) using SystemC

Program:

```
// Code your testbench here.

// Uncomment the next line for SystemC modules.

// #include <systemc>

#include <systemc>

#include <tlm>

#include <tlm_utils/simple_initiator_socket.h>

#include <tlm_utils/simple_target_socket.h>

using namespace sc_core;

using namespace tlm;

using namespace std;

// Target

struct Target : sc_module {

    tlm_utils::simple_target_socket<Target> socket;

    SC_CTOR(Target) : socket("socket") {

        socket.register_b_transport(this, &Target::b_transport);

    }

    void b_transport(tlm_generic_payload& trans, sc_time& delay) {

        int* data = (int*) trans.get_data_ptr();

        cout << "Target received: " << *data << endl;

        delay += sc_time(10, SC_NS);

    }

};

// Initiator

struct Initiator : sc_module {

    tlm_utils::simple_initiator_socket<Initiator> socket;

    SC_CTOR(Initiator) : socket("socket") {
```

```

SC_THREAD(thread_process);
}

void thread_process() {
    tlm_generic_payload trans;
    sc_time delay = SC_ZERO_TIME;
    int data = 555;
    trans.set_data_ptr(reinterpret_cast<unsigned char*>(&data));
    trans.set_data_length(sizeof(data));
    cout << "Initiator sending: " << data << endl;
    socket->b_transport(trans, delay);
    wait(delay);
}

};

int sc_main(int argc, char* argv[]) {
    Initiator initiator("initiator");
    Target target("target");
    initiator.socket.bind(target.socket);
    sc_start();
    return 0;
}

```

testbench.cpp

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2 // Uncomment the next line for SystemC modules.

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4 #include <systemc>

5 #include <tlm>

6 #include <tlm_utils/simple_initiator_socket.h>

7 #include <tlm_utils/simple_target_socket.h>

8 using namespace sc_core;

9 using namespace tlm;

10 using namespace std;

11 // Target

12 struct Target : sc_module {

13 tlm_utils::simple_target_socket<Target> socket;

14 SC_CTOR(Target) : socket("socket") {

15 socket.register_b_transport(this, &Target::b_transport);

16 }

17 void b_transport(tlm_generic_payload& trans, sc_time& delay) {

18 int* data = (int*) trans.get_data_ptr();

19 cout << "Target received: " << *data << endl;

20 delay += sc_time(10, SC_NS);

21 }

22 };

23 // Initiator

24 struct Initiator : sc_module {

design.cpp

1 //

2 //

3 //

Log

Share

[2026-08-23 15:48:13 UTC] g++ -o sim -lsystemc *.cpp && echo "Compile done. Starting run..." && ./si

Output:

Log

Share

[2026-08-23 15:48:13 UTC] g++ -o sim -lsystemc *.cpp && echo "Compile done. Starting run..." && ./sim

Compile done. Starting run...

SystemC 2.3.3-Accellera --- May 23 2020 14:33:56

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Initiator sending: 555

Target received: 555

Done